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Task Description:	This NASA Specialized Center of Research (NSCOR) aims to model the relative risk of colonic and stomach tumorigenesis for high priority space radiation beams and compare to gamma radiation where human epidemiologic data are available. It will rely primarily on mouse models of gastrointestinal (GI) cancer and human colonic epithelial cell (HCEC) approaches that have been optimized for low dose exposures. This will involve quantitative and qualitative analyses of tumors as well as normal GI epithelium after radiation. Results will be extrapolated to radiation-induced molecular changes in state-of-the-art 2D and 3D cultured human colonocytes. While limited studies at select loci, such as HRPT, have shown high energy (HZE)-induced mutations including splicing errors, frameshifts, and complex changes, we will carry out genome-wide analysis using the power of modern nextgen sequencing (NGS) to assess the spectrum of mutations in radiation-induced tumors, and also persistent genome-wide epigenomic changes in normal cells after radiation. These NGS omics findings will be integrated with transcriptomics and metabolomics results as well as events at the protein and cellular levels to develop a systems biology model for GI tumorigenesis. By assessing both quantitative in vivo tumorigenesis data and molecular events in mouse tissues and human colonocytes, space radiation-induced tumorigenesis will be compared to human epidemiologic data for low LET (linear energy transfer) radiation exposures, and approaches will be developed to model human risks for GI cancer by space radiation. We continue to employ two models of GI cancers: the APC1638N/+ model, which was generated by a targeted modification of the Apc gene at a position corresponding to aa1638. These mice develop very few adenomas in the small as well as in the large intestine but post-radiation tumor numbers show marked increases. Radiation also induced substantial gastric tumorigenesis as well in this model. The second model is the CDX2P Apc flox/+ that has one APC allele inactivated almost exclusively in the colon leading to adenomas and subsequently a low rate of adenocarcinomas in the colon that is increased post-irradiation. Data being collected from these two relevant mouse models will be used to calculate risk estimates for NASA. Since low-LET risk estimates are available for human populations, we will then be able to model the relative risk for space radiation. In particular, this ratio can then be extrapolated to human exposure to low-LET radiation to then make an estimate of risk for colorectal cancer in humans exposed to space radiation compared to the standard risks based on atomic bomb survivors and other exposure populations. In order to answer the critical question of risk for colonic tumorigenesis after acute and chronic space radiation exposures, in vitro human cell culture and animal models for colorectal cancer (CRC) initiation and progression are needed. We are investigating the effects of high-LET particle and high-energy protons in colonic epithelial cells with respect to the nature of damage and consequent tumor induction using biological cancer-related endpoints and systems biology (genomics, proteomics, and metabolomics) approaches. In addition, we propose to compare these results to the biological consequences of low dose rates and mixed fields of radiation which astronauts are likely to be continuously exposed. The studies of low- and high-LET radiation on colonocytes gain more relevance with increasing age when premalignant adenomas and other colonic lesions such as flat lesions and microscopic aberrant crypt foci become more prevalent. Pre-existing undetectable genetic or chromosomal aberrations in astronauts in addition to de novo mutations resulting from long-term irradiation may increase the chances of colonic stem cell transformation. Plans for this proposal build on our past accomplishments with GI tumorigenesis models and are focused on studies at low doses with an expanded range of NASA's high priority beams. This proposal is also focused on assessing the effects of gender, dose-rate, age, and genetic heterogeneity in mouse models. Molecular events triggered by space radiation will be investigated in depth and the mechanistic basis for persistent aberrant cell-signaling after HZE radiation will be dissected. Genome-wide NGS combined with other omics approaches will contribute to a systems biology understanding of the effects of space radiation in the intestines and stomach. Both quantitative murine tumor data and results for underlying molecular events in mouse and human GI cells will contribute to modeling of human risks for GI cancer by space travel. The overall aims of this NSCOR are summarized below: Aim 1 (Project 1). Quantitatively assess GI tumorigenesis in mouse models of GI cancer and collect samples for qualitative analysis Aim 2 (Project 2). Dissection of the signaling events and consequences in GI cells of the persistent effects of space radiation Aim 3 (Project 3): Characterization of IR(infrared)-induced neoplastic events in normal diploid human colonocytes Aim 4 (Project 4). Development of systems biology (Project 4A) and mathematical modeling (Project 4B) approaches for GI cancer risk assessment.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Gastrointestinal tumors are frequent in the U.S., and American Cancer Society estimates that there will be 140,250 new cases of colorectal cancer (CRC) with 50,630 persons predicted to die of the disease in 2018. Colorectal cancer when considered in men and women is the second leading cancer killer in the United States (http://www.cdc.gov/). Considering the high frequency of colorectal tumors in the American population, an even small increase by space radiation could have a major impact on risk estimates and planning of future manned space missions. The effect of complex DNA lesions, which are produced by HZE radiation, on intestinal tumorigenesis may provide insight into mutagenic processes affecting genome integrity and colorectal carcinogenesis.
	In 2018, we used NASA Space Radiation Laboratory (NSRL) facility for two beam runs in the spring and fall and currently, we have proposed beam time for the summer and fall of 2019. We are using the APC1638N/+ mice for quantitative and qualitative tumorigenesis data at lower doses of space radiation beams spanning a range of LET and energies as well as with mixed beam. During these beam runs, we have exposed additional mice for fulfilling statistically required numbers for different beams especially at low doses. Our previous data in APC mutant mouse model demonstrated higher intestinal tumorigenesis after exposure to 12-C, 28-Si, and 56-Fe radiation relative to gamma-rays. We also observed important qualitative differences between heavy ions and gamma-rays in both normal GI epithelial cells and tumors. While we exposed male and female mice to 5 cGy dose of gamma-rays, 16-O (325 MeV/n), 4-He (250 MeV/n), 28-Si (300 MeV/n), and 56-Fe (1000 MeV/n) as well as 10 cGy dose of 16-O (325 MeV/n) and 4-He (250 MeV/n) and presented data, we exposed additional mice and added to the data to enhance statistical power. While we started out tumorigenesis studies using 6 to 8 weeks old APC1638N/+ mice, we were also interested in comparing tumorigenesis in young vs aged mice. For aged mice, we used 20 to 24 weeks old mice which are equivalent to 30 to 50 years of human age and are within the astronauts' age range. The comparative tumorigenesis study in young vs aged was

Task Progress:

undertaken in male as well as in female mice.
After consultation with NASA, we have also carried out pilot studies to assess the potential protective effect of aspirin, which is associated with decreased frequency of human CRC. While aspirin has been reported to decrease spontaneous tumorigenesis in APC mutant mice, the experimental approach was developed to test effectiveness of aspirin in preventing/decreasing 28-Si-induced intestinal tumorigenesis in APC1638N/+ mice. We used 50 and 10 cGy 300 MeV/n 28-Si ions (LET: 70 keV/micron). We chose these doses because these are the lowest doses that demonstrated significantly higher tumorigenesis relative to sham irradiation. We have chosen two doses because 50 cGy radiation induced higher tumor relative to 10 cGy and at 10 cGy due to lower tumor number signal to noise ratio for the aspirin effect on tumorigenesis could be problematic. Aspirin was administered in diet and three doses used were equivalent to human daily doses of 75, 150, or 300 mg aspirin tablet. We used three color-coded diets containing 61.5 mg, 123 mg, 246 mg aspirin, respectively, per kg of diet. We have chosen 3 human doses of aspirin because CRC chemoprevention studies in humans have reported aspirin dose range from 75 to 300 mg/day and this dose range was also effective in removing existing adenoma. We used daily aspirin administration schedule in 6 to 8 weeks old APC1638N/+ male mice and aspirin administration was started 1 month before irradiation and was continued until the end of the study. Mice were sacrificed 150 d after radiation and intestinal and colon tumors counted. Data show that aspirin did alter the course of space radiation-induced intestinal tumorigenesis.

Task book accomplishments 2018 using CDX2P Apc flox/+ mice: Completed 3 beam fast switching experiments (30 cGy total exposure) and demonstrated that the order of beams matter (e.g., protons, then helium followed by silcon) was the most carcinogenic

1. 5 cGy of silcon did not increase carcinogenesis, nor did 20-30 cGy of proton or 5 cGy of helium. Combining 20cGy of proton and 5 cGy of helium did not increase cancer progression.

2. Using fast beam switching, 30 cGy total IR proton-helium-silicon increased cancer progression as did 27 cGy total IR (e.g., 2 cGy silicon vs 5 cGy silicon). However, 25.5 cGy of total IR (20 cGy proton, 5 cGy helium and 0.5 cGy silicon did not increase cancer progression in mouse models of cancer.

3. Using the fast beam switching and treating mice with 20 cGy protons, then 5 cGy helium and then waiting 24 hours before exposing the same mice to 5 cGy of silicon did not result in increased carcinogenesis. This suggests repair of the proton and helium damage over 24 hours was sufficient to reduce the cancer progression.

4. Pretreating mice with the countermeasure CDDO in chow for 2 days prior to the 3 beam fast switching (30 cGy total IR) was sufficient to reduce cancer progression. However, similarly feeding mice with aspirin had no effect and cancer progression.

5. Official GCR (galactic cosmic radiation) simulation: We were the first group to treat mice with an acute 50 cGy IR with 5 proton energies, 5 helium energies, and 5 heavy ions (NSRL 18B). This was followed in NSRL 18C with a 4 week protracted exposure to 50 cGy GCR simulation.

Results are in progress but initial results 100 day post IR suggest acute and protracted are equivalent in regards to chronic and persistent oxidative damage as measure by increases in lipid peroxidation. The overall goal of this project is to understand long-term effects of space radiation on normal gastrointestinal (GI) cells including stem cells. Coordinated epithelial cell migration is key to maintaining functional integrity and preventing pathological processes in gastrointestinal (GI) tissue and is essential for astronauts' health and space mission success. The GI tract with its high cell turnover is sensitive to radiation, and the effects of radiation are evident when the replacement process of cells lost during normal turnover are deregulated. In small intestine, considered a model system to study GI epithelial cell turnover, differentiated epithelial cells from crypt-base stem cells migrate along the crypt-villus axis to replace cells that shed into intestinal lumen via apoptosis at the villus tip. The current study irradiated wild type (Wt) mice (C57BL/6J; male; 6 to 8 weeks old; n=10) with 50 cGy 56-Fe (energy: 1000 MeV/nucleon; LET: 148 keV/micron) using simulated space radiation source at the NASA Space Radiation Laboratory (NSRL), Brookhaven National Laboratory (BNL). Mice were sacrificed either 7 d, 60 d, or 12 m after radiation exposure and cell migration in small intestine was assessed quantitatively and qualitatively. Iron radiation data were compared to an equal dose of gamma-rays data. We demonstrated that IEC migration along the crypt-villus axis is decreased up to 2 months after a low dose of heavy ion radiation relative to gamma-rays. Heavy ion radiation-induced decreased migration was associated with perturbation of molecular pathways involved in coordinated intestinal epithelial cell migration. We used also small intestine as a model system to delineate space radiation effects on GI stem cells. The intestine is a self-renewing tissue with an active crypt-base proliferative stem cell compartment and is sensitive to radiation exposures. We studied space radiation effects on intestinal stem cells (ISC) in male Lgr5 (Lgr5-EGFP-IRES-creERT2) mice in C57BL/6J background using low doses (<1 Gy) 56-Fe (1000 MeV/n) ions. Lgr5 mice were exposed to 0.5 Gy of 56-Fe and GFP+ Lgr5 cells were sorted from isolate intestinal cell using flow cytometry. Intracellular ROS was assessed in sorted intestinal stem cells (ISCs) using fluorescent probe CellROX and flow cytometry. Data show increased reactive oxygen species (ROS) and mitochondrial superoxide in ISCs 2-month after heavy ion radiation. We also show increased gammaH2AX and 53BP1 foci in iron irradiated samples suggesting continued presence of sub-lethal label of DNA DSB (double strand breaks) in these live sorted ISCs. We stained ISCs for PCNA, a known cell proliferation marker and showed increased PCNA staining supporting ISC proliferation in presence of chronic stress and DNA damage with potential for transformation. Future plan involves understanding whether stem cell stress is propagated to progeny and developing countermeasure strategy to block space radiation-induced persistent stem cell stress.

The goal of this project is to define the features of the tumor genomes arising in our murine models after gamma, proton, and HZE exposure. Using our well established next generation sequencing pipeline, we used a targeted to sequence the Apc and Trp53 regions to establish tumor purity and to identify the best samples for larger scale analysis. At the same time, we can identify any mutations occurring in these regions. Our data illustrate single nucleotide polymorphism analysis of the Apc region from a representative group of 12 tumors and 7 normal adjacent tissues. Allelic imbalance is observed in all tumors, but there is considerable variation in tumor purity. Over 50 tumors from the Fornace and Shay labs have been sequenced. In this fashion, we have identified the samples with the least dilution by normal DNA that are most suited to large scale sequencing. We are currently accessioning new samples in order to have sufficient material to test the hypothesis that HZE radiation has a recognizable mutation signature. From targeted sequencing, there is clear evidence that loss of the entire region is the primary mechanism of tumor initiation under all conditions. No single nucleotide mutations, small INDELS, or structural rearrangements in or flanking Apc have been identified. Excellent whole-genome copy number profiles can also be derived from this data. Our current analyses indicate at the dose levels used in this study, the tumor mutation burden from either chromosome rearrangements, aneuploidy, or small mutations

(single nucleotide or small indels) is remarkably low. We have now initiated whole genome sequencing on selected samples to validate this conclusion at maximum resolution.

It has now been firmly established that HZE radiation leads to a persistent disturbance in the architecture of the gut epithelium. To investigate mechanistic basis of this apparently epigenetic phenomenon, we are utilizing the Lgr5-GFP model established in the Fornace lab in addition to the Apc cancer prone models. The Lgr5-GFP system presents the opportunity to carry out transcriptomic analysis of cells isolated from irradiated and control mice and flow sorted into groups which reflect stem cells (GFP high), proliferating cells (GFP low), and differentiated cells (GFP negative). Cells from control, gamma, and HZE irradiated mice were obtained in triplicate, sorted by GFP expression, and analyzed by RNA-Sequencing.

Analysis of the data suggests that distinct differences in gene expression are observed at 120 days post-irradiation within the stem cell (high GFP) samples. We are currently following up on this data with pathway gene set enrichment analyses and are preparing to validate and refine our results using single cell RNA-Seq techniques. To complement the transciptome studies, we plan to use DNA for targeted bisulfite sequencing of 850,000 CpG sites distributed across the mouse genome. Ultimately, the integration of transcriptional data with DNA methylation status could provide important insights into the mechanism and consequences of the persistent effects of HZE exposure. There may be potential to develop blood based assays to detect the signature of HZE DNA methylation.

Astronauts on long-duration space exploration missions such as a planned trip to Mars and back will be exposed to a complex mixture of radiations that is not encountered on Earth. This space radiation mixture includes heavy ions, which produce difficult to repair clustered damage in cells and can be much more carcinogenic than gamma rays. In such situations, non-targeted effects caused by release of signals from cells directly hit by heavy ion track cores are likely to be the dominant mechanism producing adverse health effects from heavy ion. Reliable estimates for the endpoint-specific dose-dependent weighting function, which take non-targeted effects into consideration, are needed to predict human cancer risks from heavy ion exposure during space missions, based on available data for gamma rays. We used a mechanistically-motivated mathematical model to analyze updated mouse tumorigenesis data from the current NSCOR and to generate convenient weighting function equations and estimates for several space-relevant heavy ion types, compared with gamma rays. The updated data set now includes two additional ion types and lower, more space-relevant, doses. The model-based weighting function estimates are dose-dependent and tend to be highest (>10) at low doses, where non-targeted effect terms dominate the radiation response.

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